SwiftEdge Security & Optimizer

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Modern Problems

01

The Problem At Hand

Windows systems become slower, less organized, and less secure over time.

The tools to fix those issues are time consuming and far spread. Identifying vulnerable software can be difficult.

SwiftEdge addresses this by bringing those core functions into one simple, lightweight application.

Problem

- **Slower**
  Windows becomes much slower over time.
- **Less Organized**
  Frequent use reduces file system organization.
- **Insecure**
  As Windows OS ages, it becomes insecure or obsolete.

Opportunity

- **Unique Tool**
  A tool that combines functionality of other tools.
- **Lightweight Application**
  Many of the current applications or tools require large downloads or are resources intensive.

Solution

- **SwiftEdge**
  SwiftEdge combines tools, scripts, and other software to craft an all-in-one lightweight utility for Windows, including vulnerability scanning.

Project Goals

Improve System Performance	Enhance speed and efficiency	Enahcing speed on slower hardware. Optimizing the experience for efficiency for the user.
Automate System File Cleanup	Cleaner & Organized	Clean up hard to find system files, logs, dumps, & Recycle Bin. Less file clutter, more disk space, more organization.
Strengthen Windows Security	Secure Computing	Strengthening the security of Windows for a secure computing experience for the user.
Scan for Software Vulnerabilities	Vulnerable Software Found	Vulnerable software is found and can be updated and resolved by the user.

SwiftEdge Security & Optimizer

SwiftEdge Security & Optimizer

Performance Tweaks | File Cleanup | Security Hardening | Vulnerability Scanning

Welcome to SwiftEdge

SwiftEdge Security & Optimizer is a lightweight software application designed to enhance computer performance and security in a single, streamlined solution. The project aims to eliminate the need for multiple optimization and security tools by integrating system cleanup, performance tuning, and enhanced cybersecurity practices all into one intuitive platform.

Recommended: Create a System Restore Point before proceeding.

- [x] Create a System Restore Point before continuing

Continue

02

What is SwiftEdge Security & Optimizer?

- PowerShell Based Windows Utility
- Performance, Cleanup, Security, Vulnerability Scanning tools.
- Windows Forms Graphical User Interface Framework.
- Standalone executable package built for Windows 11.
- Optional standalone PowerShell Script.
- Requires zero third-party software.

Modules & Functionality

Performance

Power Plan
Performance Tuning
GPU Acceleration
Game Mode
Visual Optimization

Cleanup

Temporary Files & Prefetch Files
Disk Cleanup
Recycle Bin
Windows Update Cache
Logs & MiniDump Files

Security

Remote Security Settings
Disables Remote Assistance
Microsoft Defender Settings
Default Windows Firewall Rules

Vulnerability Scanner

NVD NIST CVE JSON Database
Scans all installed applications
Finds associated CVES based on application version number
Gives list of CVES & severity

Technologies

PowerShell 5.1+

Windows Forms GUI (PS Embedded)

Windows Registry

System Services (Control Panel, powercfg)

PS2EXE Compiler/Packaging

NVD NIST API CVE JSON CACHE

Testing

Manual Testing

SwiftEdge Security & Optimizer has built-in logging for module completion assurance.

Test Cases, Logging, Module Functionality, & Completion were Tested Manually in accordance to the test cases in our testing document.

Testing

Functional Requirements

Testing of SwiftEdge adhered to our projects functional requirements and aligned with our testing objective in our test-plan.

1

Test-Plan & Test-Cases

Test-plan set out test-cases to follow and ensure functionality, usability, and reliability. Based on Functional Requirements.

2

Testing Results

Testing followed our test-cases, ensuring application met or exceeded functional requirement expectations and test-case objectives.

3

Testing (cont.)

Verification Metrics

Testing focused on module completion, measurable changes in execution time, registry changes, control panel settings, scanner output, and logging behavior.

Performance

Execution Time, Setting Changes, Logging, Completion.

Cleanup

Execution Time, Temporary File Cleanup, Log Dumps, Logging, Completion.

Security

Execution Time, Service Disablement, Registry Edits, Logging, Completion.

Scanning

CVE Matching/Execution Time, Application Scanning, Report Output, Caching, Logging, Completion.

SwiftEdge Security & Optimizer

SwiftEdge Security & Optimizer

Performance Tweaks | File Cleanup | Security Hardening | Vulnerability Scanning

Performance | Cleanup | Security | Vulnerability

Vulnerability Module

Run the vulnerability scanner. Results will appear below:

```
Local JSON older than 1 day. Refreshing.
 Downloading NVD feed: https://nvd.nist.gov/feeds/json/cve/2.0/nvdcve-2.0-2026.json.gz
Download complete. Extracting...
Extracted JSON to C:\Users\Noah Huber\AppData\Local\SwiftEdge\NVD\nvdcve-2026.json
 Loading CVE JSON...
Loaded 9790 CVEs for 2026
Found 41 installed apps.
Scanning CVEs against installed apps...
==========================
  TOP 5 MOST VULNERABLE APPS
==========================
Ranked by: severity rank CVSS score

Wireshark 4.2.2 arm64  (Version: 4.2.2)
  Highest Severity: MEDIUM  Rank: 2  Score: 5.3
  Top CVEs:
    - CVE-2026-0962  [MEDIUM 5.3]
==========================
 Recommended Actions
==========================
- Use the official updater or vendor website for each listed application.
- If an app is unused, consider uninstalling it to reduce attack surface.
- After applying updates, rerun this scanner to confirm the vulnerabilities are reduced.
```

Run Vulnerability Scan

Challenges

03

Challenges

C.1 Vulnerability Scanning & NVD Rate Limits

- NVD API Rate Limits 5/30 Sec
- Live API queries limited
- Resorted to Yearly Cached JSON
- How to parse through 100K lines of the API.

C.2 PowerShell & Windows Forms

- Learning the intricacies of PowerShell & Windows Forms
- Major inspiration from ChrisTitus WinUtil
- Building an entire UI using Forms

C.3 Packaging w/PS2EXE

- Utilizing PS2EXE to compile and package an executable from multiple PowerShell scripts.
- Embedding multiple scripts at compilation does not behave like a normal compiler.
- Resorted to a single build script which creates a single PowerShell script & executable.

Future Development

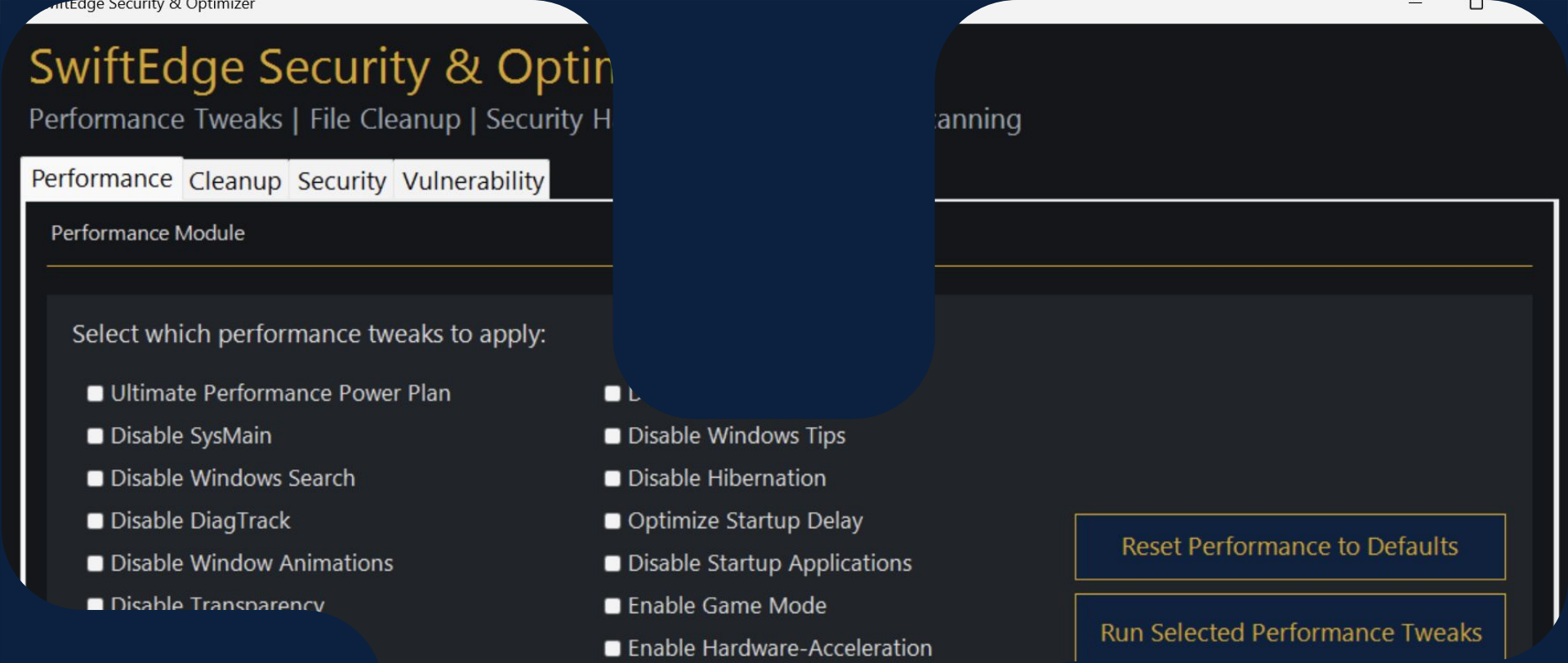
04

Future Development

STIG Profiles	Add government-backed hardening options for stronger security. Provide Additional Controls for Power Users.
Better Logging & Richer Scanner Output	Improve traceability of actions, errors, and verification of results. Add richer scanner output for scanner runs and report results. Major development could include automatically updating **Vulnerable Apps.**
Digital Certificate/ App Signing	Get digital certificate signing -> SwiftEdge is not flagged by Smartscreen or Defender. Improved executable/application distribution & packaging.

Demo & Q & A

Any Questions, Comments, Concerns?



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